

Answers: Logic Drill #7

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A

1. State the contrapositive of: “If x is prime and $x > 2$, then x is odd.”

Answer: If x is even, then x is not prime or $x \leq 2$.

Method: The contrapositive of $P \wedge Q \implies R$ is $\neg R \implies \neg P \vee \neg Q$. $\neg(\text{odd}) = \text{even}$, so: “If x is even, then x is not prime or $x \leq 2$.”

Investigate further: Is the original statement true for all positive integers x ?

2. What is the negation of: “For all real numbers x , there exists a real number y such that $x + y = 0$.”

Answer: There exists a real number x such that for all real numbers y , $x + y \neq 0$.

Method: Negating $\forall x \exists y (x + y = 0)$ yields $\exists x \forall y (x + y \neq 0)$.

Investigate further: Is the original statement true or false in $\mathbb{R}$?

3. True or false: “If a conditional statement $P \implies Q$ is true, its converse $Q \implies P$ must also be true.”

Answer: False

Method: Conditional statements and their converses are logically independent. For example, $x = 2 \implies x^2 = 4$ is true, but $x^2 = 4 \implies x = 2$ is false ($x = -2$).

Investigate further: Which related statement is logically equivalent to $P \implies Q$?

4. Which of the following is equivalent to $\neg(P \implies Q)$? (A) $\neg P \implies \neg Q$, (B) $P \wedge \neg Q$, (C) $\neg P \vee Q$.

Answer: (B) $P \wedge \neg Q$

Method: $P \implies Q \equiv \neg P \vee Q$. Taking the negation: $\neg(\neg P \vee Q) \equiv P \wedge \neg Q$ by De Morgan’s laws.

Investigate further: Under what truth values of P and Q is $P \implies Q$ false?

5. What single counterexample disproves: “ $n^2 + n + 17$ is prime for all non-negative integers n ?”

Answer: $n = 17$ (or $n = 16$)

Method: For $n = 17$, $17^2 + 17 + 17 = 17(17 + 1 + 1) = 17 \times 19 = 323$, which is composite (17×19).

Investigate further: What is the smallest non-negative integer counterexample for $n^2 + n + 41$?

6. If $A \implies B$ and $B \implies C$, what logical rule allows concluding $A \implies C$?

Answer: Hypothetical Syllogism (or Transitivity of Implication)

Method: Hypothetical Syllogism states that $(A \implies B) \wedge (B \implies C) \implies (A \implies C)$.

Investigate further: Does transitivity hold for bidirectional equivalence ($\iff$)?

7. True or false: “An assumption that leads to $0 = 1$ in a proof by contradiction proves the original premise false.”

Answer: True

Method: Assuming $\neg P$ and deriving an impossible contradiction like $0 = 1$ proves $\neg P$ is false, which forces P to be true.

Investigate further: Why is deriving $0 = 0$ insufficient in a proof by contradiction?

8. Give a counterexample to: “If $x^2 > y^2$, then $x > y$.”

Answer: $x = -3, y = 2$

Method: $(-3)^2 = 9 > 2^2 = 4$, but $-3 < 2$. The claim fails for negative numbers.

Investigate further: For which real numbers x, y is $x^2 > y^2 \implies x > y$ valid?

9. What is the base case needed for an inductive proof that $3^n > 2^n + 10$ for $n \geq 3$?

Answer: $n = 3$

Method: For $n = 3$, $3^3 = 27$ and $2^3 + 10 = 8 + 10 = 18$. Since $27 > 18$, $n = 3$ is the valid base case.

Investigate further: Why does the inequality fail for $n = 1$ and $n = 2$?

10. True or false: “The negation of ‘every student passed’ is ‘every student failed’.”

Answer: False

Method: The negation of $\forall x P(x)$ (every student passed) is $\exists x \neg P(x)$ (at least one student did not pass), not $\forall x \neg P(x)$.

Investigate further: What is the difference between negating a universal quantifier vs swapping it to an opposite universal?

Section B

1. Prove by contrapositive: “If n^2 is an odd integer, then n is an odd integer.”

Answer: Proof by contrapositive

Method: The contrapositive is: “If n is even, then n^2 is even.”

Proof: If n is even, $n = 2k$ for some integer k . Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$. Since $2k^2 \in \mathbb{Z}$, n^2 is even. Since the contrapositive is true, the original statement is true.

Investigate further: How does this compare to a direct proof of the same claim?

2. Find all integer counterexamples in $-10 \leq x \leq 10$ to: “If $x^2 > 4$, then $x > 2$.”

Answer: $x \in \{-10, -9, -8, -7, -6, -5, -4, -3\}$

Method: $x^2 > 4 \iff x > 2 \text{ or } x < -2$. The integers in $[-10, 10]$ satisfying $x < -2$ are $-10, -9, -8, -7, -6, -5, -4, -3$. For all these values, $x^2 > 4$ is true, but $x > 2$ is false.

Investigate further: How many integer counterexamples exist in total?

3. Prove by contradiction: “ $\sqrt{3}$ is an irrational number.”

Answer: Proof by contradiction

Method: Suppose toward a contradiction that $\sqrt{3} = p/q$ where $p, q \in \mathbb{Z}^+$ and $\gcd(p, q) = 1$. Squaring gives $3 = p^2/q^2 \implies p^2 = 3q^2$. Thus $3 \mid p^2 \implies 3 \mid p$, so $p = 3k$. Then $(3k)^2 = 3q^2 \implies 9k^2 = 3q^2 \implies q^2 = 3k^2 \implies 3 \mid q$. Thus $3 \mid \gcd(p, q)$, contradicting $\gcd(p, q) = 1$.

Investigate further: Does this proof extend to $\sqrt{p}$ for any prime p ?

4. Classify the condition “ $x = 2$ ” as necessary, sufficient, both, or neither for “ $x^2 - 5x + 6 = 0$ ”.

Answer: Sufficient but not necessary

Method: Sufficient: If $x = 2$, $2^2 - 5(2) + 6 = 0$, so $x^2 - 5x + 6 = 0$ holds. Not necessary: $x = 3$ also satisfies $x^2 - 5x + 6 = 0$, so $x = 2$ is not required.

Investigate further: Which set of values is necessary and sufficient for $x^2 - 5x + 6 = 0$?

5. Classify the condition “ $x^2 - 5x + 6 = 0$ ” as necessary, sufficient, both, or neither for “ $x = 2$ ”.

Answer: Necessary but not sufficient

Method: Necessary: For $x = 2$ to hold, $x^2 - 5x + 6 = 0$ must hold. Not sufficient: $x^2 - 5x + 6 = 0$ allows $x = 3$, so it does not guarantee $x = 2$.

Investigate further: How does swapping hypothesis and conclusion swap necessary and sufficient roles?

6. Prove by induction: $\sum_{i=1}^n 2^i = 2^{n+1} - 2$ for all positive integers n .

Answer: Proof by induction

Method: Base case $n = 1$: $\sum_{i=1}^1 2^i = 2$; $2^{1+1} - 2 = 4 - 2 = 2$, true.

Inductive step: Assume $\sum_{i=1}^k 2^i = 2^{k+1} - 2$. Then $\sum_{i=1}^{k+1} 2^i = (2^{k+1} - 2) + 2^{k+1} = 2 \cdot 2^{k+1} - 2 = 2^{k+2} - 2$. By induction, the formula holds for all $n \geq 1$.

Investigate further: What is the sum formula for a finite geometric series $\sum_{i=0}^{n-1} r^i$?

7. Give a counterexample to: “If a, b are irrational, then a^b is irrational.”

Answer: $a = \sqrt{2}^{\sqrt{2}}, b = \sqrt{2}$

Method: If $\sqrt{2}^{\sqrt{2}}$ is rational, it is a counterexample ($a = \sqrt{2}^{\sqrt{2}}, b = \sqrt{2}$ both irrational). If $\sqrt{2}^{\sqrt{2}}$ is

irrational, take $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$, both irrational. Then $a^b = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{\sqrt{2} \times \sqrt{2}} = \sqrt{2}^2 = 2 \in \mathbb{Q}$, a rational number.

Investigate further: This non-constructive proof establishes the existence of an irrational power of an irrational being rational without knowing which case is true!

8. Prove by contradiction: “If n is an integer and $n^3 + 5$ is odd, then n is even.”

Answer: Proof by contradiction

Method: Assume toward a contradiction that n is odd, so $n = 2k + 1$. Then $n^3 + 5 = (2k + 1)^3 + 5 = 8k^3 + 12k^2 + 6k + 1 + 5 = 2(4k^3 + 6k^2 + 3k + 3)$, which is even. This contradicts $n^3 + 5$ being odd. Thus n must be even.

Investigate further: Could this also be proved directly by showing $n^3 + 5$ odd $\implies n^3$ even $\implies n$ even?

9. State the negation of: “For every $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x - a| < \delta$, then $|f(x) - L| < \varepsilon$.”

Answer: There exists $\varepsilon > 0$ such that for all $\delta > 0$, there exists x with $|x - a| < \delta$ and $|f(x) - L| \geq \varepsilon$.

Method: Negating $\forall \varepsilon > 0 \exists \delta > 0 (P \implies Q)$ yields $\exists \varepsilon > 0 \forall \delta > 0 (P \wedge \neg Q)$. Here P is $|x - a| < \delta$ and $\neg Q$ is $|f(x) - L| \geq \varepsilon$.

Investigate further: This is the formal definition that $\lim_{x \rightarrow a} f(x) \neq L$ in analysis.

10. Prove by induction that $7^n - 1$ is divisible by 6 for all positive integers n .

Answer: Proof by induction

Method: Base case $n = 1$: $7^1 - 1 = 6$, divisible by 6.

Inductive step: Assume $7^k - 1 = 6m$. Then $7^{k+1} - 1 = 7 \cdot 7^k - 1 = 7(6m + 1) - 1 = 42m + 6 = 6(7m + 1)$, which is divisible by 6. By induction, $6 \mid (7^n - 1)$ for all $n \geq 1$.

Investigate further: How can this be proved using modular arithmetic ($7 \equiv 1 \pmod{6}$)?

Section C

1. For real numbers x, y , consider: (after TMUA 2022 Paper 2 Q10)

I. $\forall x \exists y (x + y = 0)$

II. $\exists y \forall x (x + y = 0)$

III. $\forall x \forall y (x + y = 0)$

Which of I, II, III are true?

Answer: B) I only

Method: I: For any given x , choosing $y = -x$ gives $x + (-x) = 0$, so I is true.

II: Requires a single fixed y such that $x + y = 0$ for ALL x . If $x = 0 \implies y = 0$, but then $1 + 0 \neq 0$, so II is false.

III: False (e.g. $x = 1, y = 1 \implies 1 + 1 = 2 \neq 0$). Thus I only is true.

Investigate further: Why does order of quantifiers ($\forall x \exists y$ vs $\exists y \forall x$) change the meaning completely?

2. Determine the complete set of real values k for which the statement: “For all real x , if $x > k$,

then $x^2 > k$ is true. (after TMUA 2022 Paper 2 Q9)

Answer: A) $k \geq 1$ or $k \leq 0$

Method: Case $k \leq 0$: If $x > k$, then $x^2 \geq 0 \geq k$. If $k = 0$, $x > 0 \implies x^2 > 0 = k$. True for $k \leq 0$.

Case $0 < k < 1$: Take $x \in (k, 1)$. Then $x > k$, but $x^2 < x < 1$. Specifically for $x = (k+1)/2$, $x > k$ holds, but $x^2 < k$ holds, so statement fails.

Case $k \geq 1$: If $x > k \geq 1$, then $x^2 > x > k$, true.

Thus the statement holds for $k \leq 0$ or $k \geq 1$.

Investigate further: Verify $k = 0.5$: Take $x = 0.6 > 0.5$, then $x^2 = 0.36 < 0.5$, counterexample!

3. Consider positive integers a, b . Statement P : “ $a + b$ is even.” Statement Q : “ $a^2 + b^2$ is even.” Which of the following is true?

Answer: C) P is necessary and sufficient for Q .

Method: $a + b$ is even $\iff a, b$ have the same parity $\iff a^2, b^2$ have the same parity $\iff a^2 + b^2$ is even. Thus $P \iff Q$, meaning P is necessary and sufficient for Q .

Investigate further: What if the statement was $a + b$ is odd vs $a^2 + b^2$ is odd?

4. A student attempts to prove: “There are no integer solutions to $x^2 + y^2 = 7$.”

Answer: A) The proof is completely valid and correct.

Method: Squares modulo 4 are $0^2 \equiv 0, 1^2 \equiv 1, 2^2 \equiv 0, 3^2 \equiv 1$. Thus $x^2 \pmod{4} \in \{0, 1\}$ and $y^2 \pmod{4} \in \{0, 1\}$. Their sum $(x^2 + y^2) \pmod{4} \in \{0, 1, 2\}$. But $7 = 4(1) + 3 \equiv 3 \pmod{4}$. Since $3 \notin \{0, 1, 2\}$, no integer solutions exist. The proof is sound.

Investigate further: Which other integers cannot be written as the sum of two squares?

5. For a real number x , let P : “ $x^2 - 4 = 0$ ” and Q : “ $x = 2$ ”. Which of the following statements is true?

Answer: B) $Q \implies P$ is true, but $P \implies Q$ is false.

Method: $Q \implies P$: If $x = 2$, $2^2 - 4 = 0$, true. $P \implies Q$: $x^2 - 4 = 0 \implies x = 2$ or $x = -2$, so $x = -2$ is a counterexample to $P \implies Q$. Thus $Q \implies P$ is true, but $P \implies Q$ is false.

Investigate further: Is $x = 2$ necessary or sufficient for $x^2 - 4 = 0$?

6. Consider the claim: “For every positive integer n , $n^3 + 2n$ is divisible by 3.” Which proof technique is most direct?

Answer: A) Factorise $n^3 + 2n = n(n^2 + 2) = n(n^2 - 1 + 3) = n(n-1)(n+1) + 3n$, where $(n-1)n(n+1)$ is divisible by 3.

Method: $n^3 + 2n = n^3 - n + 3n = (n-1)n(n+1) + 3n$. The product of 3 consecutive integers $(n-1)n(n+1)$ is divisible by 3, and $3n$ is divisible by 3, so their sum is divisible by 3 directly.

Investigate further: Can this also be proved by testing $n \equiv 0, 1, 2 \pmod{3}$?

7. What is the negation of “All prime numbers are odd or equal to 2”?

Answer: B) There exists a prime number that is even and not equal to 2.

Method: Negating $\forall p, (P(p) \vee Q(p))$ gives $\exists p, (\neg P(p) \wedge \neg Q(p))$. $\neg(\text{odd}) = \text{even}$, and $\neg(p = 2) = p \neq 2$. Thus: “There exists a prime number that is even and not equal to 2.”

Investigate further: Why does De Morgan’s law switch ‘or’ to ‘and’ under negation?

8. Let S be a set of n distinct positive integers. Consider the statement: “If $n \geq 5$, then S contains two numbers whose difference is a multiple of 4.” Is this statement true or false? (after SMC 2017 Q18)

Answer: A) True, by the Pigeonhole Principle (4 possible remainders mod 4).

Method: When an integer is divided by 4, there are 4 possible remainders: 0, 1, 2, 3. By the Pigeonhole Principle, if $n \geq 5$ integers are selected, at least two must have the same remainder modulo 4. If $a \equiv b \pmod{4}$, then $a - b \equiv 0 \pmod{4}$, meaning $a - b$ is a multiple of 4.

Investigate further: How many integers must be chosen to guarantee two whose difference is a multiple of k ?

Section D

1. Prove by contradiction: “There do not exist integers x, y such that $x^2 - 5y^2 = 3$.” (after BMO1 2015 Q1)

Answer: Proof by contradiction

Method: Assume toward a contradiction that there exist integers x, y such that $x^2 - 5y^2 = 3$. Taking this equation modulo 5, we get $x^2 \equiv 3 \pmod{5}$. The squares modulo 5 are $0^2 \equiv 0, 1^2 \equiv 1, 2^2 \equiv 4, 3^2 \equiv 4, 4^2 \equiv 1 \pmod{5}$. Thus $x^2 \pmod{5} \in \{0, 1, 4\}$. Since $3 \notin \{0, 1, 4\}$, $x^2 \equiv 3 \pmod{5}$ has no integer solutions. Contradiction.

Investigate further: What are the quadratic residues modulo 5?

2. A student claims: “If a, b, c are positive integers such that $a \mid bc$, then $a \mid b$ or $a \mid c$.” (after TMUA 2019 Paper 2 Q7)

Answer: (a) $a = 6, b = 2, c = 3$ ($6 \mid 6$ but $6 \nmid 2$ and $6 \nmid 3$). (b) $\gcd(a, b) = 1$ (or a is prime); Euclid’s Lemma.

Method: (a) Take $a = 6, b = 2, c = 3$. $bc = 6$, so $6 \mid 6$ is true. However, $6 \nmid 2$ and $6 \nmid 3$, disproving the claim.

(b) The minimal condition is $\gcd(a, b) = 1$ (or requiring a to be prime). The result is Euclid’s Lemma: If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.

Investigate further: Why does Euclid’s Lemma require coprime factors or prime divisors?

3. Prove by mathematical induction that for all positive integers n , $\sum_{i=1}^n i \cdot i! = (n+1)! - 1$.

Answer: Proof by induction

Method: Base case $n = 1$: $\sum_{i=1}^1 i \cdot i! = 1 \cdot 1! = 1$; $(1+1)! - 1 = 2! - 1 = 1$, true.

Inductive step: Assume $\sum_{i=1}^k i \cdot i! = (k+1)! - 1$. Then $\sum_{i=1}^{k+1} i \cdot i! = [(k+1)! - 1] + (k+1) \cdot (k+1)! = (k+1)![1 + (k+1)] - 1 = (k+1)!(k+2) - 1 = (k+2)! - 1 = ((k+1)+1)! - 1$. By induction, the identity holds for all $n \geq 1$.

Investigate further: How does this identity relate to telescopic sums using $i \cdot i! = (i+1)! - i!$?

4. Consider three statements about a real function $f : \mathbb{R} \rightarrow \mathbb{R}$: (after TMUA 2023 Paper 2 Q12)

- I. If $f(x) > 0$ for all $x > 0$, then $f(1) > 0$.
- II. If $f(1) > 0$, then $f(x) > 0$ for all $x > 0$.
- III. If $f(x) \leq 0$ for some $x > 0$, then $f(1) \leq 0$.

Answer: I and III are logically equivalent to each other.

Method: Statement I has the form $P \implies Q$, where P is “ $f(x) > 0$ for all $x > 0$ ” and Q is “ $f(1) > 0$ ”. Statement III is the contrapositive of I: $\neg Q$ is “ $f(1) \leq 0$ ”, and $\neg P$ is “ $\exists x > 0, f(x) \leq 0$ ”. Since a statement and its contrapositive are logically equivalent, $I \iff III$. Statement II is the converse of I, which is not logically equivalent.

Investigate further: Why does a statement always have the exact same truth value as its contrapositive?

5. Prove by contradiction that there are infinitely many prime numbers (Euclid’s Theorem).

Answer: Proof by contradiction

Method: Assume toward a contradiction that there are only finitely many prime numbers, $p_1, p_2, \dots, p_k$. Consider the integer $N = p_1 p_2 \dots p_k + 1$. Since $N > 1$, N must have at least one prime factor p . Since $p_1, \dots, p_k$ are all the primes that exist, p must be one of them, so $p \mid N$. But $p \mid (p_1 p_2 \dots p_k)$, so $p \mid (N - p_1 p_2 \dots p_k) \implies p \mid 1$. But no prime divides 1, contradiction! Thus, infinitely many primes exist.

Investigate further: Does Euclid’s construction $N = p_1 p_2 \dots p_k + 1$ always yield a prime number itself?